

Midterm Assignment #1

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BSIT 2-3

COMP 009 - OOP

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Introduction to Java

①. What is Java programming language? Discuss its key features and advantages.

> Java is a high-level programming language used to develop applications such as desktop programs, web systems, and Android apps.

> Key Features:

- Object-oriented (uses class and objects to organize code)
- Platform-independent (write once, run anywhere)
- secure, and has automatic memory management

> Advantages:

- Can run on different operating systems
- Widely used in companies and large systems
- Good for beginners because of its structured syntax

②. Explain the concept of platform independence in Java.

How does Java achieve platform independence?

> Platform independence means a Java program can run on different systems like Windows, Mac, or Linux without rewriting the code. Java does this by converting the program into bytecode, which runs on the Java Virtual Machine (JVM) instead of directly on the computer. As long as the device has JVM, the program will work.

3. Discuss the main components of a Java program and its roles.

- > Class - main container of the program
- > Main Method - starting point where the program execution begins
- > Variables - store data or values used in the program
- > Methods - blocks of code that perform specific tasks/actions
- > Import statements - used to include built-in Java libraries

4. What are the differences between JDK, JRE, and JVM in Java?

- > **JDK** (Java Development Kit) - for creating Java programs, includes compiler (javac) and development tools
- > **JRE** (Java Runtime Environment) - for running Java programs, contains JVM and standard libraries
- > **JVM** (Java Virtual Machine) - executes the bytecode and converts it to machine code, allowing Java programs to run on any system

5. Explain the difference between interpreted and compiled languages. Is Java an interpreted or compiled language?

- > A compiled language translates the whole program into machine code before running, while an interpreted language runs code line by line. Java is both because it is first compiled into bytecode, then the JVM interprets and executes it.